

CUSTOM GPTS & CLAUDE ARTIFACTS

O'Reilly Live Training — No-code AI Tools Cheatsheet

Custom GPTs · Claude Artifacts · Claude Desktop MCPs · 15 Prompt Templates · Claude vs ChatGPT

1. CUSTOM GPTS — WHAT & WHY

4 Custom GPT Use Cases

- **Translators of Complexity** — simplify domain knowledge for non-experts
- **Tools** — automate repetitive tasks (extraction, formatting, scheduling)
- **Agents** — autonomous task execution with minimal supervision
- **Collaborators** — work alongside you as dedicated thought partners

When to Build a Custom GPT

- Writing too many edits to system messages
- Complex tasks need multiple specialized prompts
- Need consistent behavior across sessions
- Require specialized domain knowledge grounded
- **Tip:** Search GPT store first; build only for unique needs

CUSTOM GPT BUILDER STEPS

1

Create

chatgpt.com/gpts →
Create

2

Name & Describe

Set name + short
description

3

Instructions

Role · Rules · Output
format

4

Knowledge

Upload PDF, DOCX,
TXT files

5

Capabilities

Web · DALL·E · Code
Interpreter

6

Actions

External APIs via
OpenAPI schema

7

Test

Preview panel — iterate
& refine

8

Save & Share

Publish to GPT Store or
share link

CUSTOM GPT EXAMPLES

Italian Tutor

Simulates a 20-year-experience Italian tutor for translation and conversation practice.

- **Translation mode:** Italian text + contextual explanation
- **Scenario mode:** Italian description + English prompt for user response
- **Feedback:** Short response critique after each user attempt
- **Knowledge:** Uploaded PDF grounds scenarios to course material
- **Instructions structure:** ROLE → PROCESS → LIMITATIONS → OUTPUT CONSTRAINTS → EXAMPLES

Data Extractor

Medical research paper extraction engine — turns PDFs into structured tables.

- **Step 1:** User uploads PDF + extraction instructions
- **Step 2:** GPT confirms column structure: *Title / Author / ...*
- **Step 3:** On confirmation, extract data into copy-ready table
- **Fallback:** Outputs "Can't extract data requested" if unavailable
- **Instructions structure:** ROLE → PROCESS (3 steps) → OUTPUT CONSTRAINTS → LIMITATIONS

2. CLAUDE ARTIFACTS

5 Artifact Content Types

- **Documents** — Markdown, Plain Text (reports, notes)
- **Code snippets** — Python, JS, SQL, and more
- **Websites** — single-page HTML apps (interactive, shareable)
- **SVG images & diagrams** — charts, flowcharts, logos
- **Interactive React** — buttons, forms, data visualizations

Artifact Features

- Dedicated preview window alongside chat
- Fully interactive and editable inline
- Version history — iterate without losing prior versions
- Export and sharing capabilities
- Save to Claude Projects for organization
- Collaborative editing

Regular Chat

- Inline text in the conversation
- Primarily text-based output
- Static, non-editable
- Best for shorter content
- Limited visualization
- No version history

Artifacts

- Dedicated preview window
- Multiple output formats
- Fully editable inline
- Best for substantial content (>15 lines)
- Rich visualization (React, SVG, HTML)
- Full version history

Claude Artifact Example — Data Visualizer

Single HTML/JS app that takes JSON or CSV data and produces interactive charts (bar, line, scatter, etc.) with an **"AI Data Generate"** button that calls Claude API to create synthetic data. Built with a single Claude prompt requesting a self-contained single-page application.

3. CLAUDE DESKTOP + MCPS (MODEL CONTEXT PROTOCOL)

What MCPs Add to Claude

- Open standard by Anthropic for AI-to-tool connections
- Access to local **file system** (read/write files)
- Connect to **databases** — query and write data
- Connect to **external APIs** (GitHub, Slack, calendars)
- Control **web browsers** and apps
- Enables fully personal AI assistants

MCP Setup (Claude Desktop)

1. Install Claude Desktop app (claude.ai/download)
2. Edit `claude_desktop_config.json`
3. Add MCP server entries with command + args
4. Restart Claude Desktop
5. Verify tools appear in conversation UI

Key MCPs: filesystem · browser · GitHub · Slack · databases

4. PROMPT TEMPLATES — 15 REUSABLE TEMPLATES

Template Variable Syntax

- `{VARIABLE_NAME}` — simple placeholder; replace with your value
- `{VAR|opt1, opt2, opt3}` — multi-option; choose one or use your own
- `{VAR|description}` — single hint/example of what to provide

Example: `{FORMAT|PDF, HTML, DOCX}` → pick "PDF" or write "LaTeX"

How to Use a Template (4 Steps)

1. Choose template from `prompts/prompt-templates/`
2. Scan for all `{VARIABLE}` placeholders
3. Replace each with your specific value
4. Customize further — templates are starting points, not rigid rules

15 Templates by Category

Category	Templates (3 shown per row)
Creative & Design (3)	create-brand-assets · create-custom-webpage · turn-inspiration-to-design-plans
Content Creation (3)	create-company-newsletter · create-digital-recipe-cards · create-custom-course-materials
Business & Productivity (3)	create-sales-reports · create-interactive-pdf-forms · create-a-process-flowchart
Research & Analysis (4)	research-data-analysis · plan-literature-review · turn-text-threads-to-researched-notes · turn-research-into-presentations
Travel & Planning (2)	compare-travel-destinations · create-daily-travel-itinerary

Prompt Engineering Best Practices

- **Structure:** Role + Context + Task + Format — the 4 elements of a strong prompt
- **Few-shot:** Include 1–3 input/output examples to guide the AI's style
- **Chain prompts:** Break complex work into multi-step prompt sequences
- **Iterate:** AI assistants are forgiving — start simple and refine based on output
- **Combine:** Mix elements from multiple templates for hybrid use cases

5. CLAUDE VS CHATGPT — KEY DIFFERENCES

Quick Comparison		
Feature	Claude (Anthropic)	ChatGPT (OpenAI)
Context Window	200K standard · 1M-token beta	Up to 400K tokens (API)
Image Generation	No native image gen	gpt-image-2 (DALL·E retired)
Voice	Basic (mobile beta, English only)	Advanced Voice Mode + video/screen share
Default Model	Sonnet 5 (balanced everyday default)	GPT-5.5 Instant
Custom AI Configs	Projects + system prompts	Custom GPTs (full builder + GPT Store)
Interactive Output	Artifacts (HTML, React, SVG, code)	Canvas (code + writing editor)
Coding	Claude Code + Opus 4.8	Canvas + GPT-5.5 Instant
Computer Control	Computer Use (API beta)	Operator agent

Claude Models (mid-2026)

Table 5: frontier / most-capable tier

Opus 4.8: deep reasoning + agentic coding

Sonnet 5: balanced everyday default

Haiku 4.5: fast, budget-friendly

ChatGPT Models (mid-2026)

GPT-5.5 Instant: current default model

GPT-5.6 (Sol / Terra / Luna): limited preview, not yet GA

Pricing & Quotas

Plan names, prices, and message quotas change frequently — verify each vendor's current pricing page before relying on specifics in class.

6. DECISION FRAMEWORK — WHICH TOOL FOR WHAT?

Custom GPTs vs. Claude Artifacts

- **Custom GPT:** Persistent specialized ChatGPT config; saves instructions + knowledge files for reuse across sessions
- **Artifact:** Claude's dedicated output window for substantial, interactive, shareable content in a single session
- Use Custom GPT when you need a *deployable tool* others can use
- Use Artifacts when you need *interactive output* to iterate on in-session

Choose Your Path

- Need consistent ChatGPT behavior + domain knowledge? → **Custom GPT**
- Need interactive/shareable Claude output (app, chart, doc)? → **Artifact**
- Need Claude to access files, apps, or APIs? → **Claude Desktop + MCP**
- Need a reusable prompt structure anyone can fill in? → **Prompt Template**
- Need long-document analysis (>100K tokens)? → **Claude** (200K–1M context)
- Need image generation built-in? → **ChatGPT** (gpt-image-2)

KEY TAKEAWAYS

- **Custom GPTs:** Build specialized ChatGPT configurations with role, rules, knowledge files, and API actions — reusable by anyone
- **Claude Artifacts:** Get interactive, editable outputs (HTML apps, React, SVG, code) with version history and sharing built in
- **Claude Desktop + MCPs:** Connect Claude to your local files, databases, and APIs to build a personal AI assistant
- **Prompt Templates:** Use `{VARIABLE}` syntax for reusable, customizable prompts across 5 domains (15 templates total)
- **Platform choice:** Claude wins on context length + coding; ChatGPT wins on image gen + voice — both excel at core AI tasks
- **Iteration is key:** AI assistants are forgiving — start simple, use templates as starting points, and refine based on output